HDF5 Audit Report

[NN]

Very abstract...

hello

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Revision History

Version Number	Date	Comments
v1	July 4, 2026	Initial draft released for internal review.
v2	August 1, 2026	Public release.

Contents

1 Introduction

1.1 Purpose

1.2 Scope

Out of scope:

- PHDF5
- HDFView
- HSDS
- AI

1.2.1 Safety

1.2.2 Security

1.2.3 Privacy

2 Motivation

2.1 Problem Statement

2.2 Goals

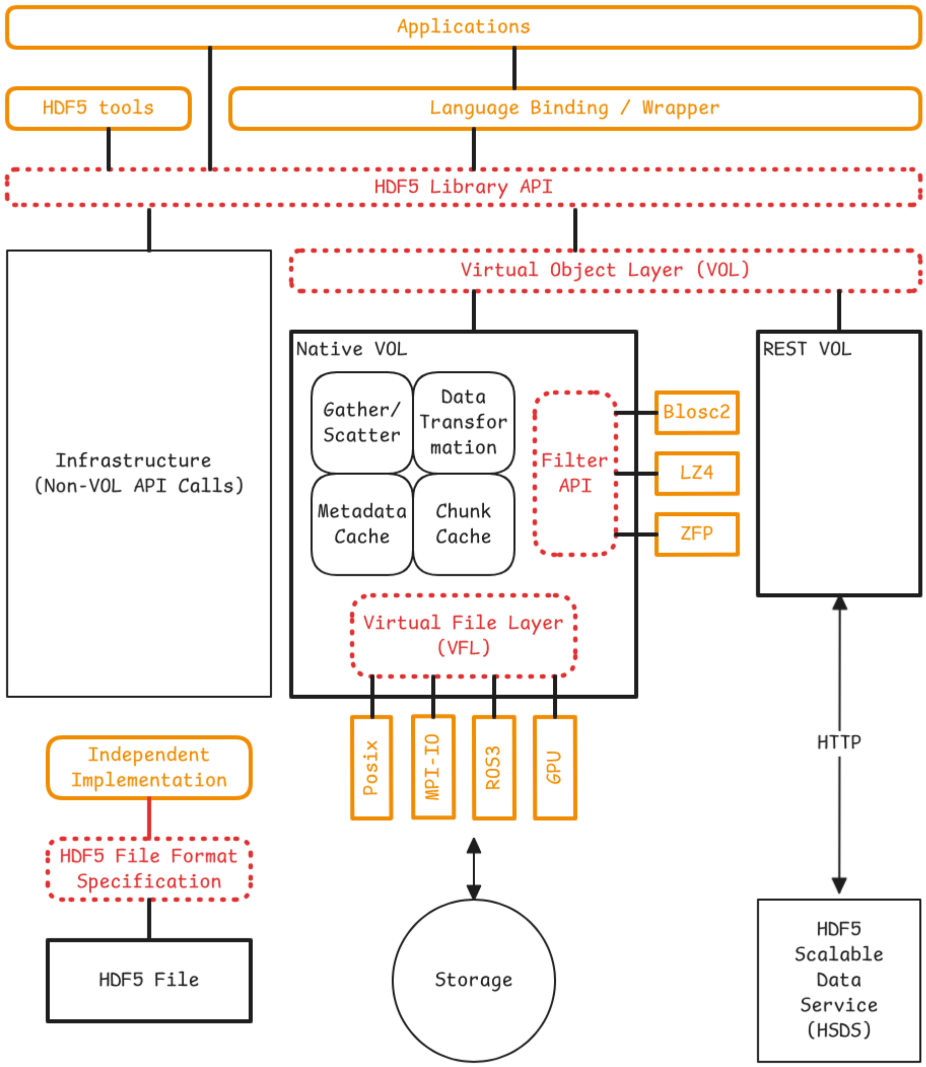

Figure 1 – HDF5 vulnerability sites (Interfaces: red – Supply chain: orange)

3 HDF5 Core Library and File Format

3.1 Accidents Waiting to Happen – HDF5 Safety

- Disk full
- File corruption
- Long term data accessibility and interpretability

3.2 Verify and Verify – HDF5 Security

- Integer arithmetic
- Running with scissors

3.3 Footprints, Fingerprints, Controls – HDF5 Privacy

- It's all in the open
- Middle ground
- All-out encryption

3.4 HDF5 Tools

3.5 Audit Findings and Mitigation Priorities

4 HDF5 Extensions

4.1 Overview

4.2 Virtual Object Layer (VOL) Connectors

4.3 Filters Plugins

4.4 Virtual File Drivers (VFD)

4.5 Audit Findings and Mitigation Priorities

5 HDF5 Wrappers and Language Bindings

5.1 Overview

Complete:

- C#
- Java
- Python
- JavaScript
- Rust

Partials...

5.2 Python

- CVE-2025-995 [8]
- CVE-2026-0897 [9]
- CVE-2026-1669 [10]
- CVE-2026-2492 [7]

5.3 Java

5.4 Audit Findings and Mitigation Priorities

6 Independent HDF5 Implementations

6.1 Overview

- HDF5 library playing footsie with the file format spec: GitHub issue 6409 [1]

6.2 Audit Findings and Mitigation Priorities

7 HDF5 in the Software Supply Chain

- HDF5 2.0.0 source-artifact integrity check [11]
- SBOM

7.1 Audit Findings and Mitigation Priorities

8 Long-term Data Accessibility and Interpretability

8.1 Audit Findings and Mitigation Priorities

9 Resources

- HDF SSP SIG [6]
- Tools
  - HDF5 CVE test suite [3]
  - A machine-readable specification of the HDF5 file format. [2]
  - HDF5 extension templates [4]
  - HDF5 plugins [5]
- S2D2

References

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[2] The HDF Group. *A machine-readable specification of the HDF5 file format*. 2026. URL: https://github.com/HDFGroup/hdf5-pickles. (accessed: 21.05.2026).

[3] The HDF Group. *HDF5 CVE Test Suite*. 2026. URL: https://github.com/HDFGroup/cve_hdf5. (accessed: 21.05.2026).

[4] The HDF Group. *HDF5 extension skeletons, literally*. 2026. URL: https://github.com/HDFGroup/hdf5-boneyard. (accessed: 21.05.2026).

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[7] Zero Day Initiative. *CVE-2026-2492*. 2026. URL: https://www.cve.org/CVERecord?id=CVE-2026-2492. (accessed: 21.05.2026).

[8] Google LLC. *CVE-2025-9905*. 2025. URL: https://www.cve.org/CVERecord?id=CVE-2025-99057. (accessed: 21.05.2026).

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